

1. Setting Up the Environment

Log into **Grace**, navigate to your project directory, and run the following commands sequentially. The final command submits your job to the cluster scheduler:

None

```
module purge
module load Anaconda3
conda tos accept --override-channels --channel
https://repo.anaconda.com/pkgs/main
conda tos accept --override-channels --channel
https://repo.anaconda.com/pkgs/r
conda create -y -n yolo_env python=3.10
conda activate yolo_env
conda install -c conda-forge ultralytics -y
conda install -c conda-forge pytorch torchvision ultralytics -y
conda install -c conda-forge albumentations -y
pip install simsimd
sbatch runJob.sh
```

2. Required Directory & File Structure

Your project workspace on the cluster must be organized as follows to ensure YOLO can locate your images, labels, and configuration files correctly:

None

```
/scratch/user/<you>/
├── YOLOTrainer.py
├── yolo26x.pt
├── runJob.sh
├── dataset/
│   ├── train/
│   │   ├── images/ (--- training image files)
│   │   └── labels/ (--- training .txt label files)
│   ├── val/
│   │   ├── images/ (--- validation image files)
│   │   └── labels/ (--- validation .txt label files)
│   └── test/
│       ├── images/ (--- test image files)
│       └── labels/ (--- test .txt label files)
└── specs.yaml
```

3. Choosing and Downloading Alternative Models

If you want to train a different model variant, you can download alternative weight files directly into your environment.

- Refer to the [Ultralytics YOLO Documentation](#) to see available model sizes and naming conventions.
- Use `wget` on the login node to download a model (example for `yolo26x.pt`):

None

```
wget
```

```
https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26x.pt
```

4. Important Data & Configuration Notes

- **Bounding Box Label Format:**
- Labels must be formatted for object detection as text files where each line follows this structure:
- `[class_id] [x-center] [y-center] [width] [height]` (all coordinates must be normalized values between `0.0` and `1.0`).

Data Augmentations Warning:

- Ensure that any applied augmentations do not push bounding box coordinates outside the valid `0–1` boundary range. If coordinates become illegal (negative or `>1.0`), YOLO will automatically discard those images.

Learn more about available options in the [YOLO Data Augmentation Guide](#).

Configuring Classes:

To update or change your class definitions and dataset paths, modify your `specs.yaml` file.

5. Locating Training Outputs & Results

Once your training job completes successfully, your output files and evaluation metrics will be organized automatically under your run directories:

- **Model Weights & Artifacts:** Found in `runs/detect/train/` (look inside the `weights/` subfolder for `best.pt`).
- **Validation Statistics & Plots:** Found in `runs/detect/val/`.